Chap. 1. Signals and Systems

. Signals

. Systems

. Sinusoids, Exponential

. Continuous, Discrete

. Periodic, Aperiodic signals

. Frequency

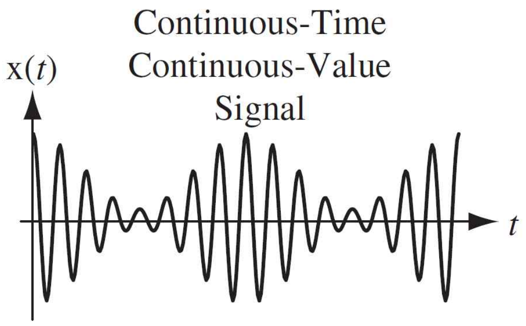
. Operation on signals

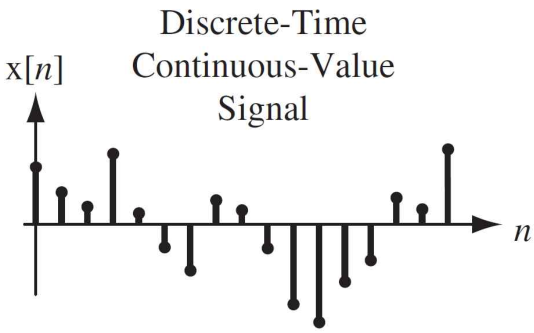
. Step function, Impulse function

. Linear, Time-invariant, Causality,
Stability, Memory

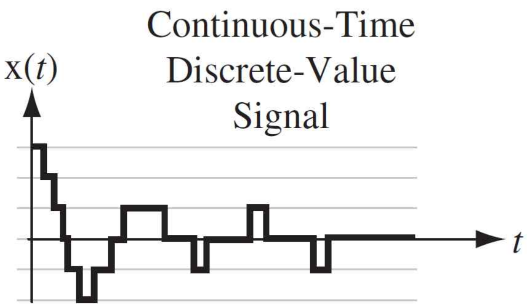
§1.1 Continuous-time and Discrete-time Signals

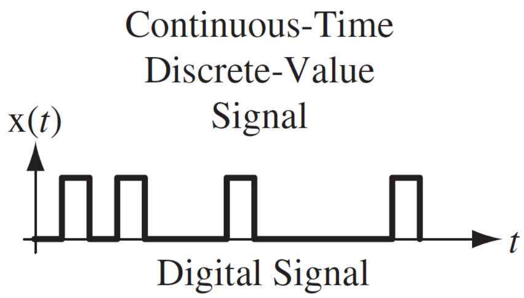
※ Examples of continuous-time and discrete-time signals



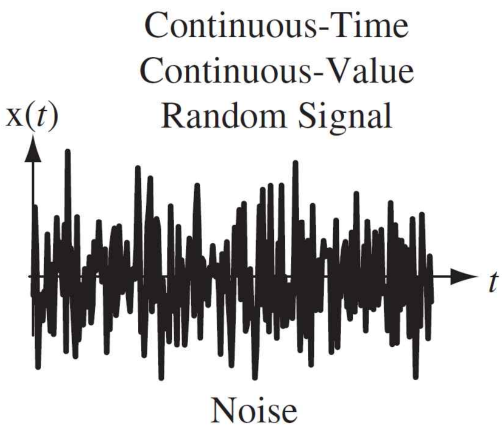


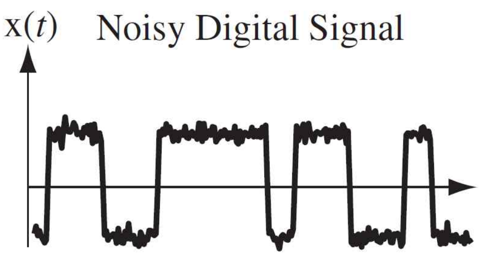
※ Examples of continuous-time, discrete-value signals





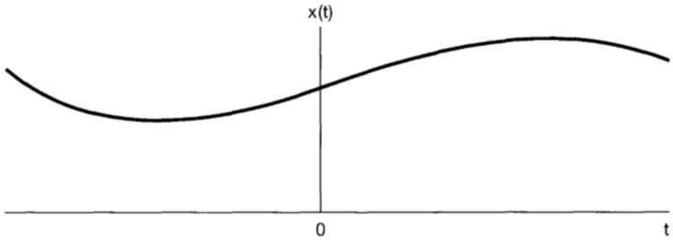
※ Examples of noise and a noisy digital signal





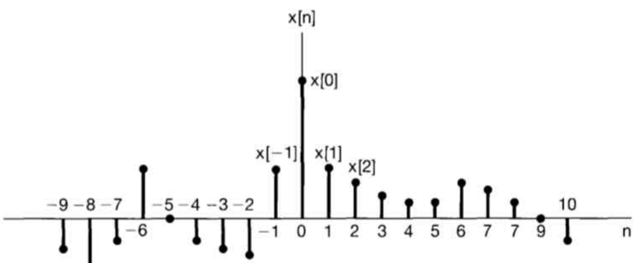
- <u>Graphical representations</u>

  · Example of the continuous-time signal $x(t)$



(a)

  · Example of the discrete-time signal $x[n]$

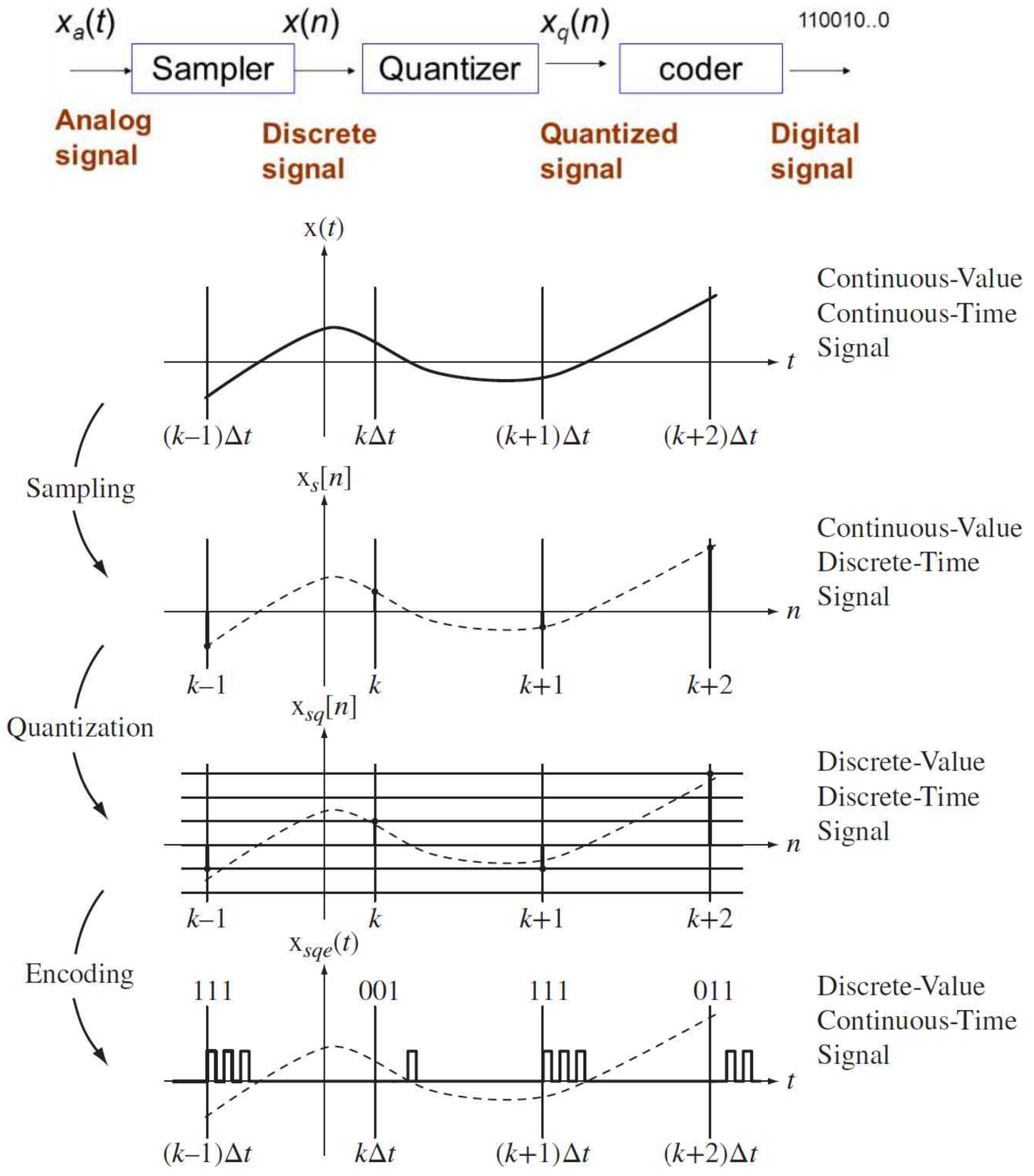


(b)

  · The discrete-time signal $x[n]$ is defined ONLY for integer values of the independent variable, n.
  · That is, $x[n]$ is a discrete-time SEQUENCE.
  · A discrete-time signal, $x[n]$, can be obtained from the SAMPLING of a continuous-time signal, $x(t)$.
  · Unit of 't' in $x(t)$: second (i.e. a real time)
    Unit of 'n' in $x[n]$: NONE (i.e. a dimensionless quantity)

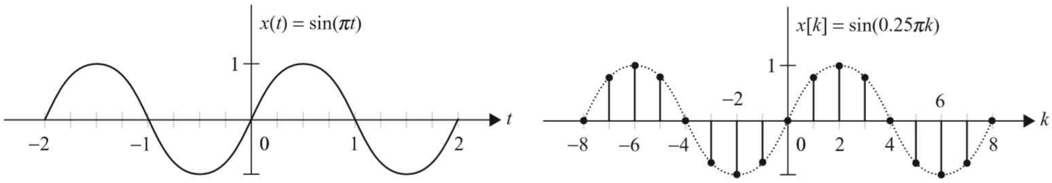
※ How can we make a digital signal? ---- A/D converter

※ Basic parts of an A/D Converter (Pulse Code Modulation)



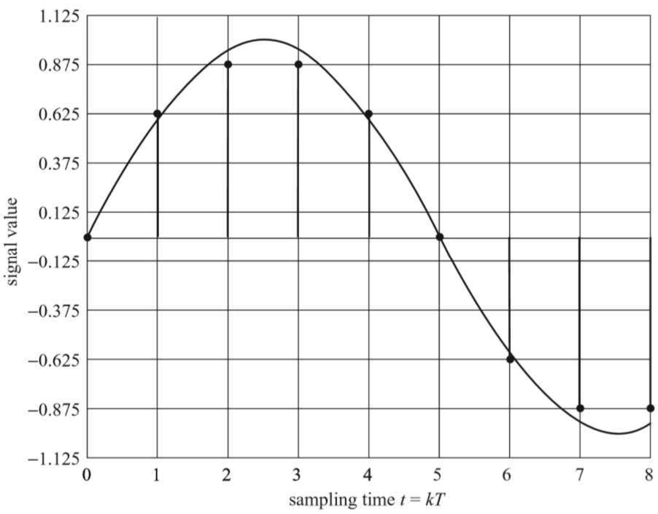
(Example) [Sampling]

$$x(t) = \sin(\pi t) \text{ and } x[kT] = \sin(\pi k T) \ \{T = 0.25\,[\text{sec}]\}$$



$$\begin{aligned}
&x[-8] = x(-8T) = \sin(-2\pi) = 0, && x[1] = x(T) = \sin(0.25\pi) = \frac{1}{\sqrt{2}},\\
&x[-7] = x(-7T) = \sin(-1.75\pi) = \frac{1}{\sqrt{2}}, && x[2] = x(2T) = \sin(0.5\pi) = 1,\\
&x[-6] = x(-6T) = \sin(-1.5\pi) = 1, && x[3] = x(3T) = \sin(0.75\pi) = \frac{1}{\sqrt{2}},\\
&x[-5] = x(-5T) = \sin(-1.25\pi) = \frac{1}{\sqrt{2}}, && x[4] = x(4T) = \sin(\pi) = 0,\\
&x[-4] = x(-4T) = \sin(-\pi) = 0, && x[5] = x(5T) = \sin(1.25\pi) = -\frac{1}{\sqrt{2}},\\
&x[-3] = x(-3T) = \sin(-0.75\pi) = -\frac{1}{\sqrt{2}}, && x[6] = x(6T) = \sin(1.5\pi) = -1,\\
&x[-2] = x(-2T) = \sin(-0.5\pi) = -1, && x[7] = x(7T) = \sin(1.75\pi) = -\frac{1}{\sqrt{2}},\\
&x[-1] = x(-T) = \sin(-0.25\pi) = -\frac{1}{\sqrt{2}}, && x[8] = x(8T) = \sin(2\pi) = 0,\\
&x[0] = x(0) = \sin(0) = 0.
\end{aligned}$$

※ Quantized Digital Signal

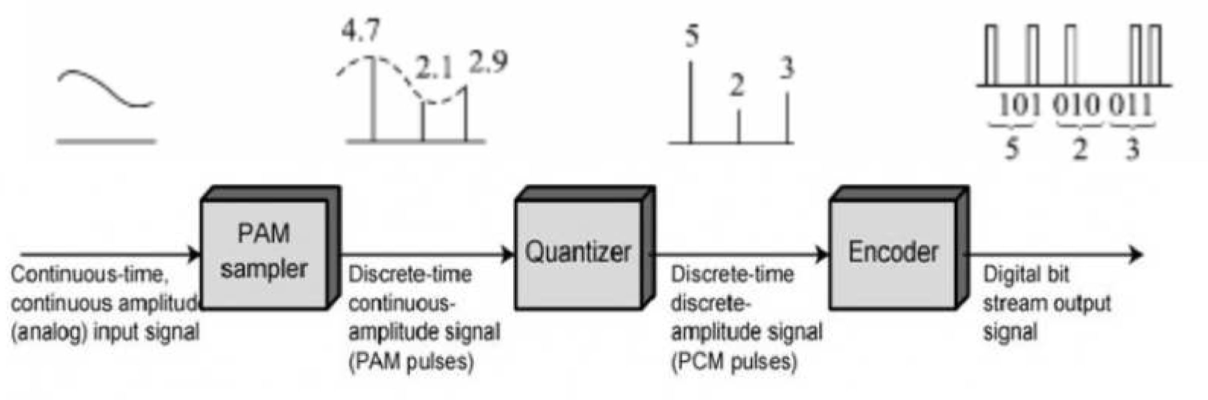


(Example) Sampling/Quantization/Encoding

(PCM: Pulse Code Modulation)

continuous-time,
continuous-value
signal
(Analog signal)

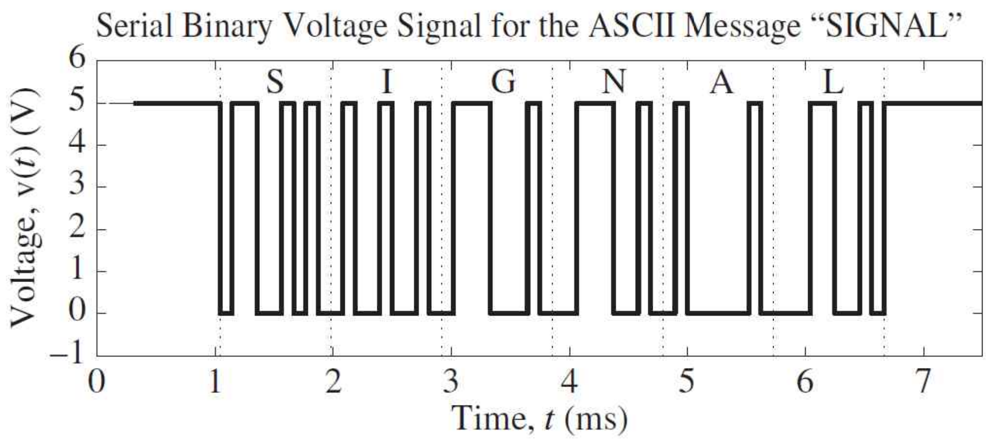
continuous-time,
discrete-value
signal
(Digital signal)



(Example) A coded signal

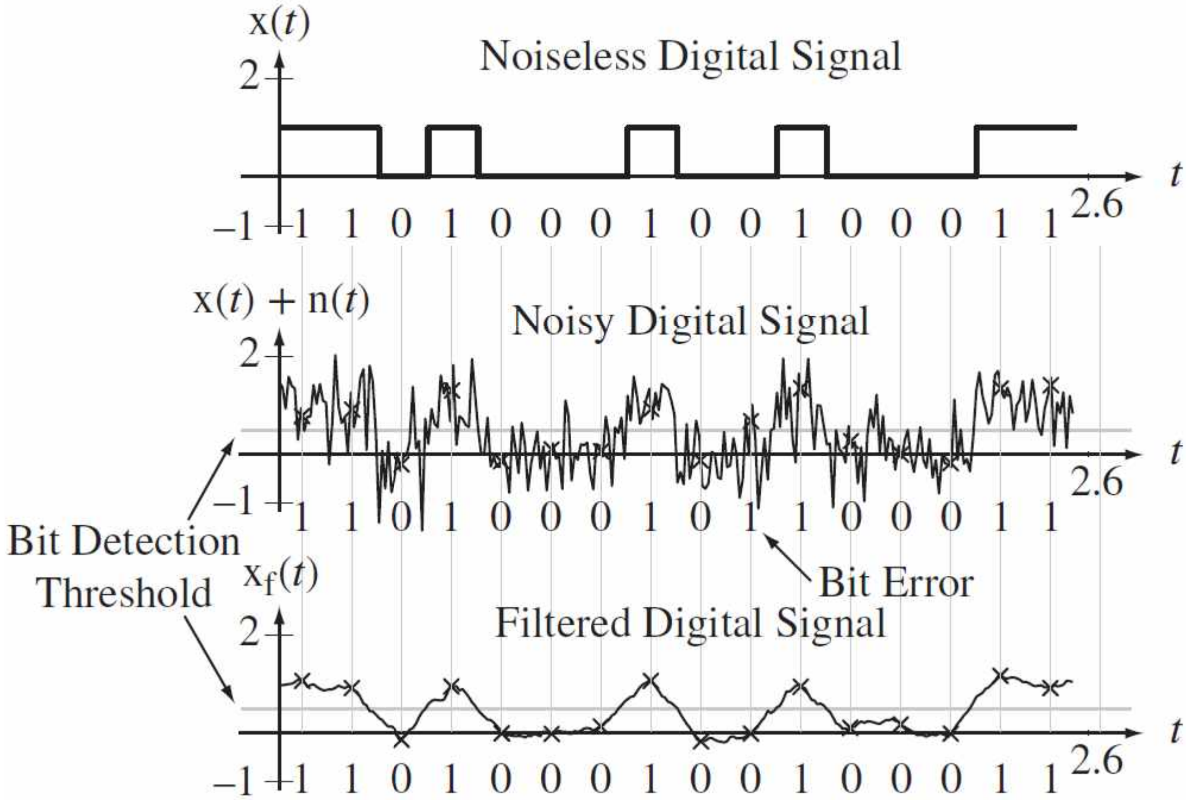
Serial Binary ASCII-encoded voltage signal for the word SIGNAL

(※ signal from keyboard to computer CPU)

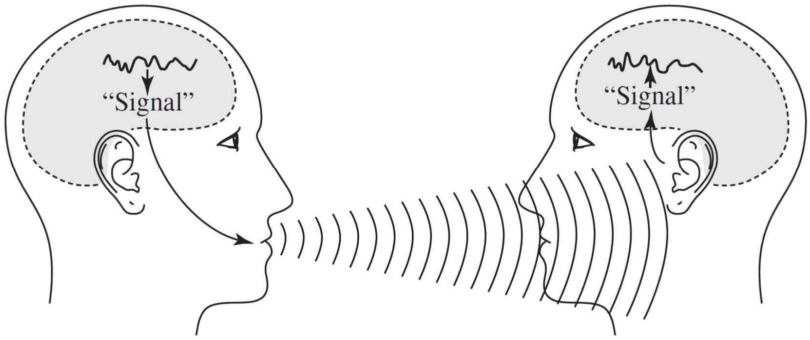


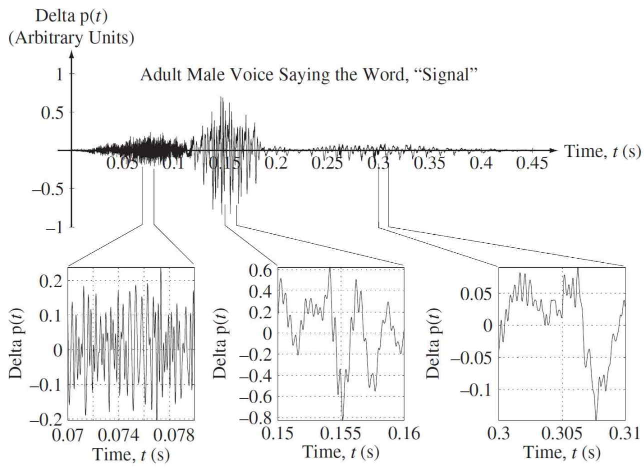
※ Regeneration of the signal from a noisy digital signal

(Low-Pass Filter: LPF)



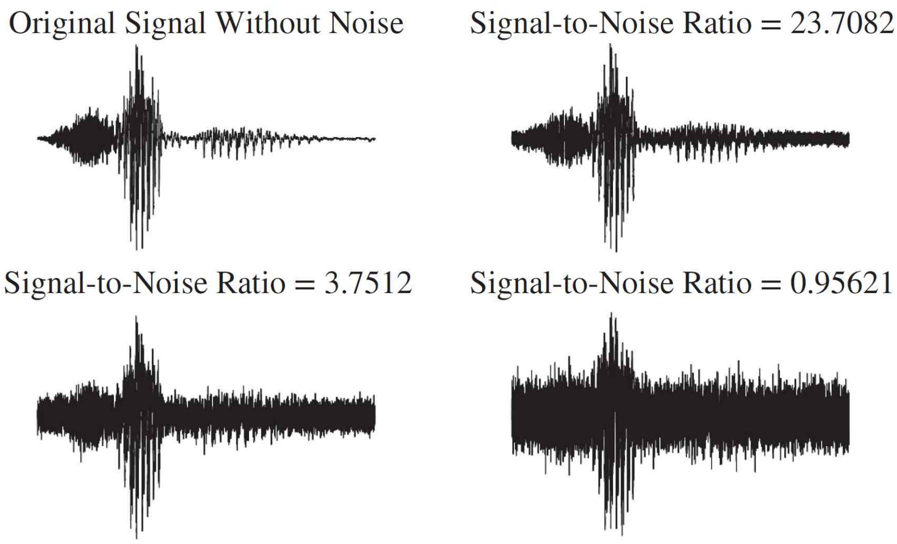
※ Communication between two people





※ Sound of the word "signal" with different levels of noise added

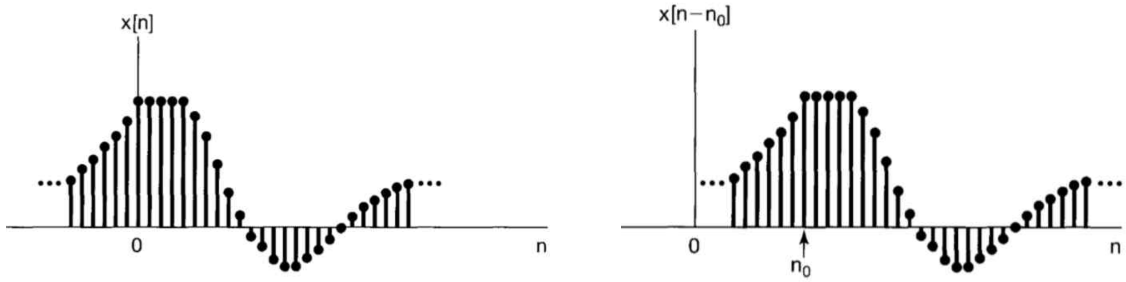
(signal from human to human)



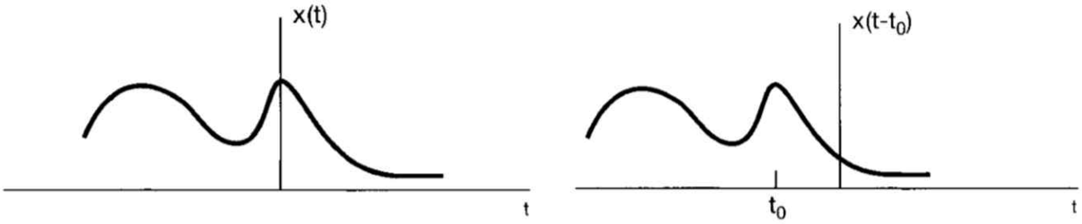
§1.2 Transformations of the Independent Variable

- Time shift

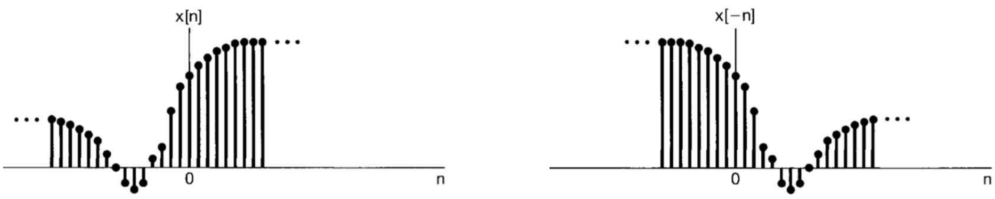
  Example: a delayed version of $x[n]$, when $n_0 > 0$

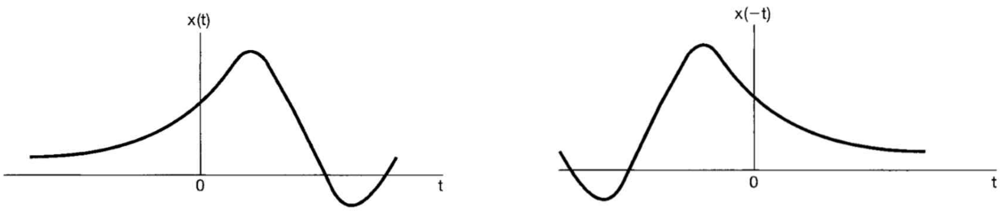


Example: an advanced version of $x(t)$, when $t_0 < 0$



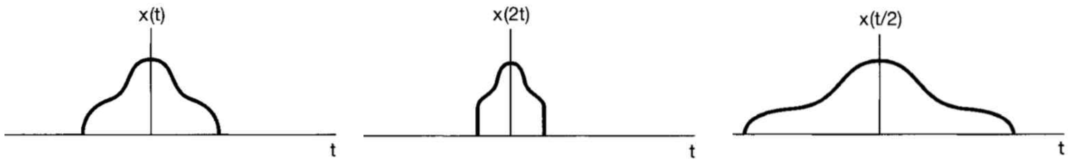
- Reflection





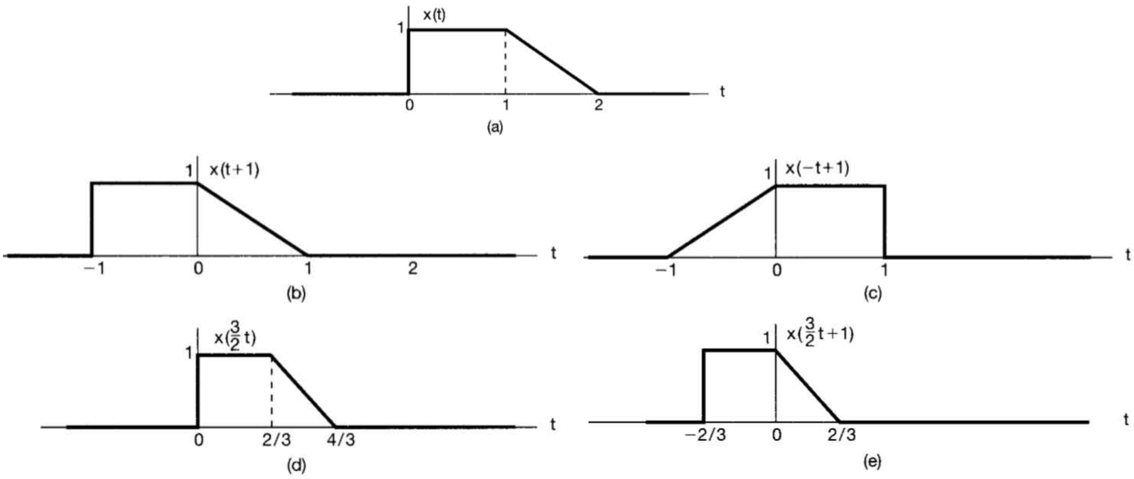
- Scaling

  Example: continuous-time signals related by time scaling



Example 1.1

(a) $x(t)$, (b) $x(t+1)$, (c) $x(-t+1)$, (d) $x(\frac{3}{2}t)$, (e) $x(\frac{3}{2}t+1)$

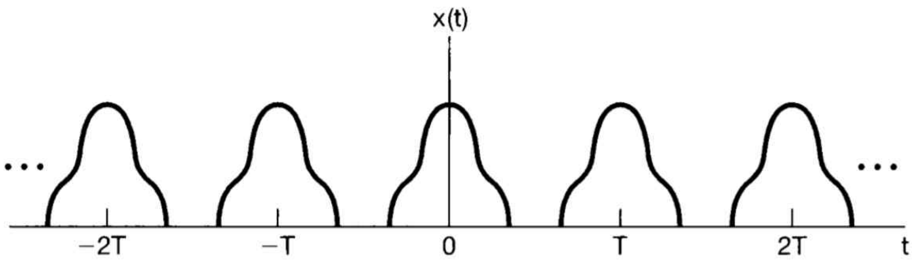


- Periodic signals
  - Continuous-time periodic signals

    $x(t) = x(t+T)$: periodic with T, $2T$, $3T$, ...

    T_o : fundamental period (the smallest positive value of periods)

(Example)

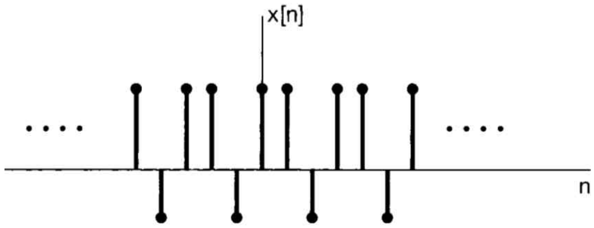


· Discrete-time periodic signals

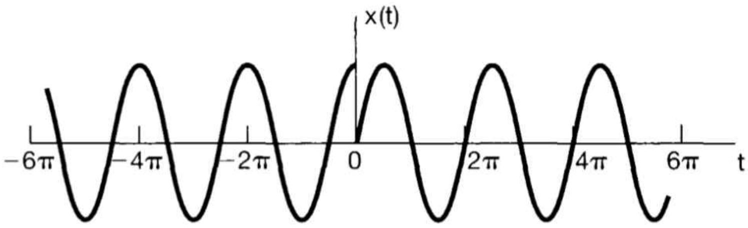
$x[n] = x[n+N]$: periodic with N, $2N$, $3N$, ...

N_o : fundamental period (the smallest positive value of periods)

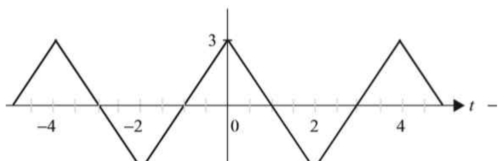
(Example)

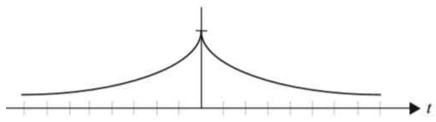


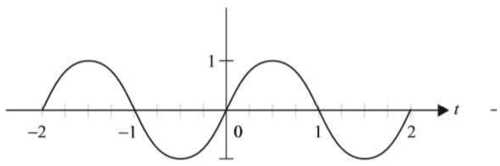
• Example 1.4 $x(t)$ is periodic?

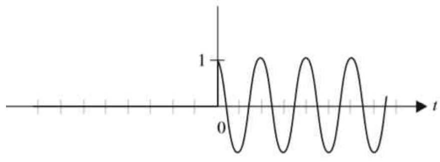


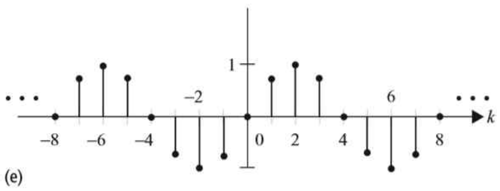
(Examples)

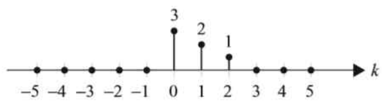
<table>
<tr><th>periodic signals</th><th>nonperiodic signals</th></tr>
<tr><td>

(a)</td><td>

(b)</td></tr>
<tr><td>

(c)</td><td>

(d)</td></tr>
<tr><td>

(e)</td><td>

(f)</td></tr>
</table>

(Example) Periodic or nonperiodic? Why?

(i) $f[k] = \sin(\pi k/12 + \pi/4)$; (ii) $g[k] = \cos(3\pi k/10 + \theta)$; (iii) $h[k] = \cos(0.5k + \phi)$; (iv) $p[k] = e^{j(7\pi k/8+\theta)}$.	(i) periodic (ii) periodic (iii) nonperiodic (iv) periodic

- Even and Odd Signals

even signals

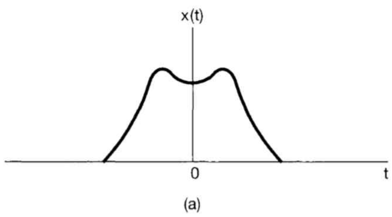
$$x(-t) = x(t) \quad \text{or} \quad x[-n] = x[n]$$

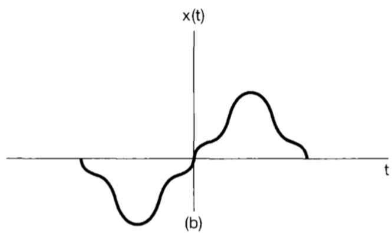
odd signals

$$x(-t) = -x(t) \quad \text{or} \quad x[-n] = -x[n]$$

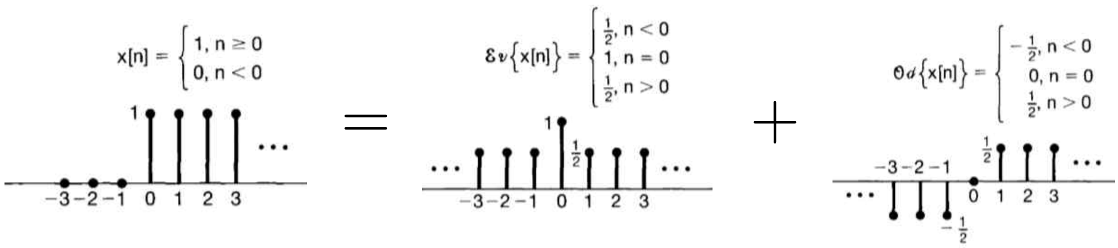
(Example)

an even continuous-time signal　　an odd continuous-time signal





※ Any signal can be broken into a sum of two signals, one of which is even and one of which is odd.

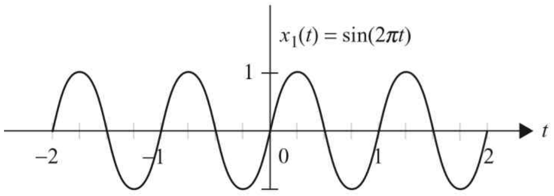
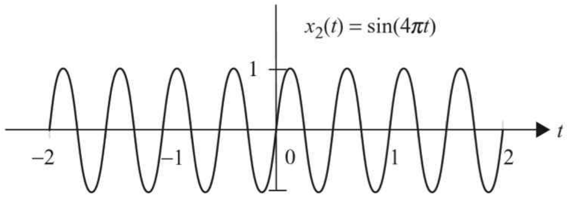


a function　　even function　　odd function

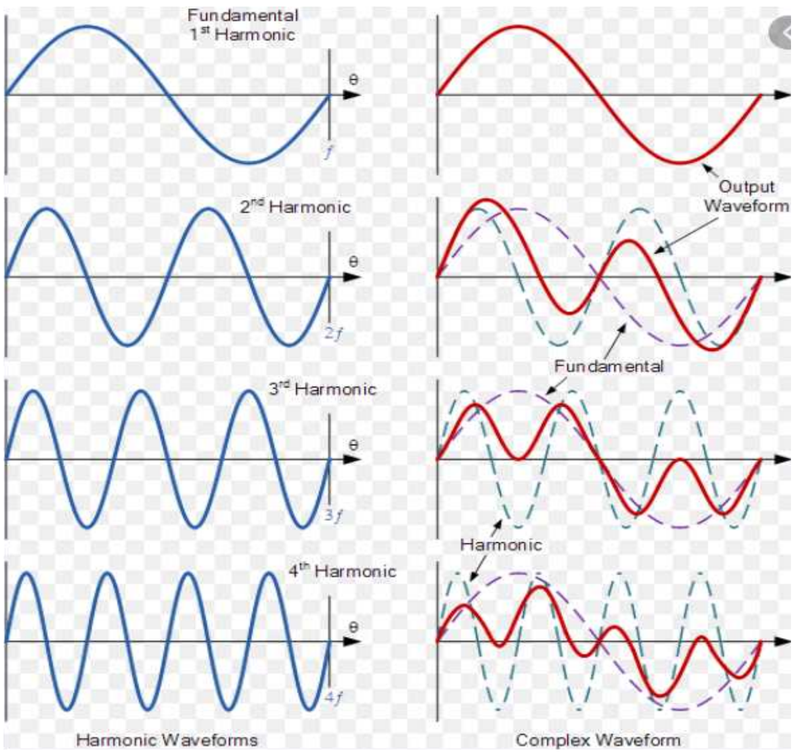
- Harmonics

  fundamental sinusoidal waveform : $x(t) = \sin(\omega_0 t + \theta)$

  the mth harmonic of $x(t)$: $x_m(t) = \sin(m\,\omega_0 t + \theta)$

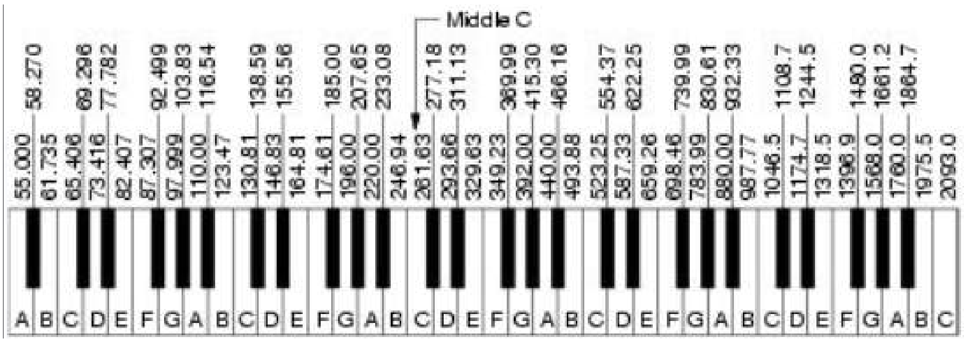
fundamental waveform	2nd harmonic waveform
	

(Example)



- Piano Keys

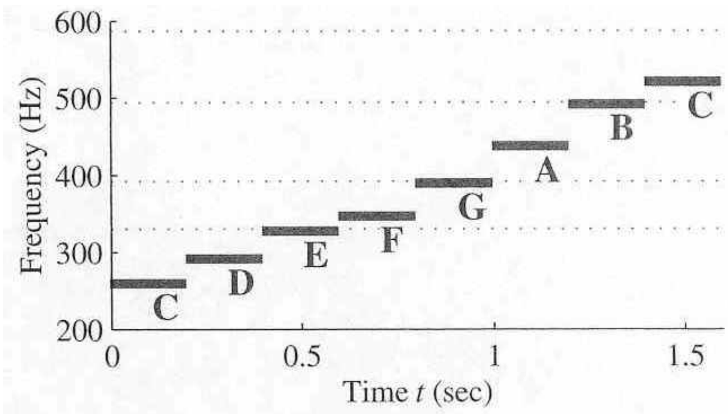
Piano keys can be numbered from 1 to 99.
Middle C is key 40. A-440 is key 49.



Middle C	D	E	F	G	A	B	C
262 Hz	294	330	349	392	440	494	523

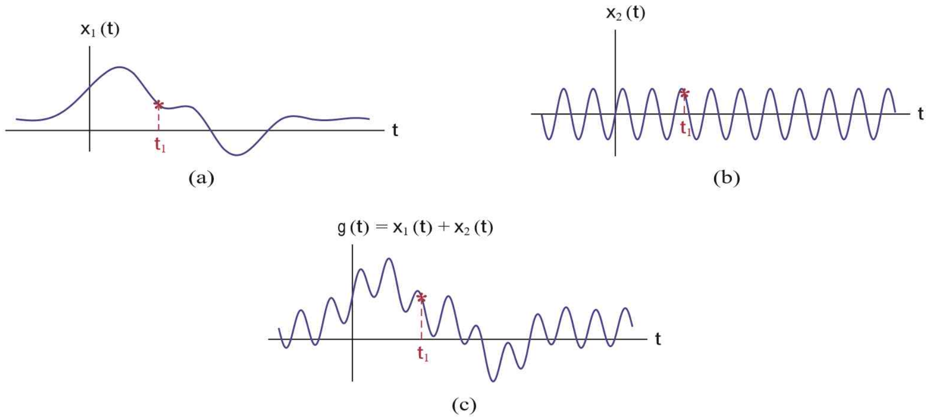
※ whole tone, semitone

※ Spectrogram

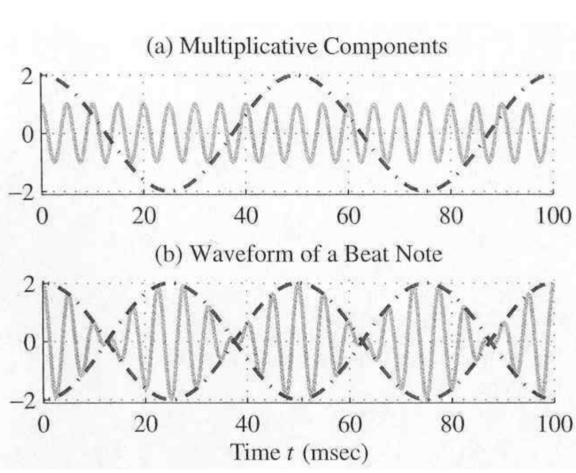
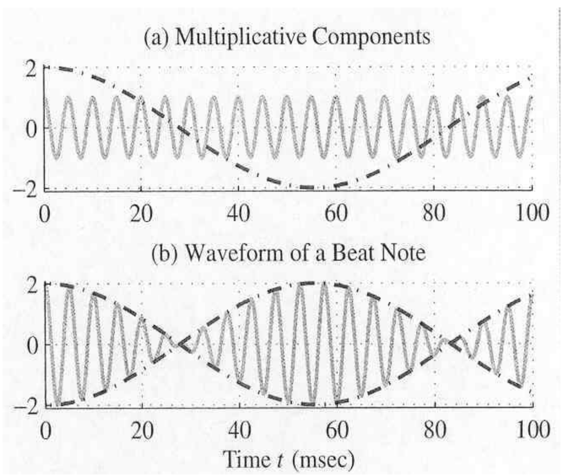


https://dspfirst.gatech.edu/

- Addition of two signals
  (Example) $g(t) = x_1(t) + x_2(t)$



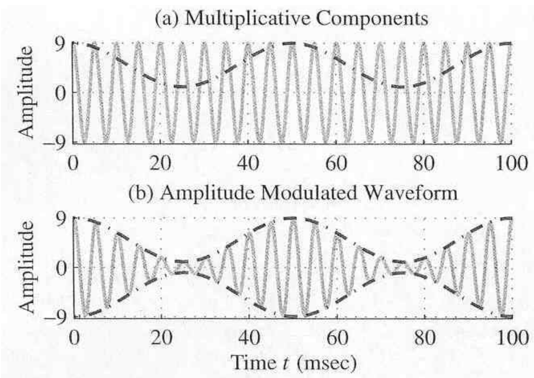
- Multiplication of two signals
  (Example) $g(t) = x_1(t) \times x_2(t)$

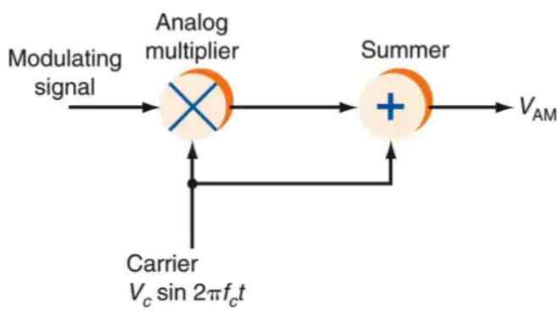
200Hz and 20Hz	200Hz and 9Hz
	

- Multiplication of two sinusoids including dc

  (Example) AM/FM radio

Waveform of Amplitude Modulation ([dc+20Hz] and 200Hz)





modulating signal	$x(t) = \cos \omega t$
carrier	$x_c(t) = \cos \omega_c t$
modulated signal	$\mathrm{V_{AM}} = [\, x(t) \times x_c(t) \,] + x_c(t)$ $= [\, \cos \omega t \times \cos \omega_c t \,] + \cos \omega_c t$ $= [\, \cos \omega t + 1 \,] \times \cos \omega_c t$

§1.3 Exponential and Sinusoidal Signals

§1.3.1 Continuous-time Complex Exponential & Sinusoidal Signals

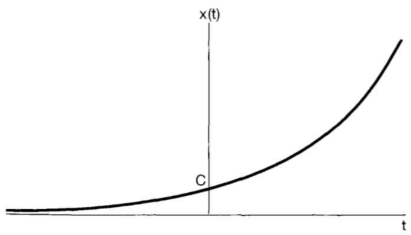
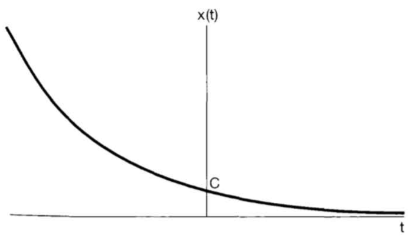
- Continuous-time complex exponential signal

$$x(t) = Ce^{at} \qquad (1.20)$$

C and a are, in general, complex numbers.

(1) Real Exponential Signals

If C and a are real, $x(t)$ is called a real exponential.

$a > 0$ (a growing exponential)	$a < 0$ (a decaying exponential)
	
- chain reactions in atomic explosions - chemical reactions	- the process of radioactive decay - the responses of RC circuits - damped mechanical systems

If $a = 0$, then $x(t) =$ constant.

(2) Periodic Complex Exponential and Sinusoidal Signals

If C is real and a is purely imaginary, $x(t)$ is sinusoidal.

Consider a periodic complex exponential signal

$$x(t) = e^{j\omega_0 t}$$

This signal will be periodic with period T , if

$$e^{j\omega_0 t} = e^{j\omega_0 (t+T)} = e^{j\omega_0 t} e^{j\omega_0 T}$$

For periodicity, we must have

$$e^{j\omega_0 T} = 1$$

If $\omega_0 = 0$, then $x(t) = 1$ [i.e., periodic for any value of T]

If $\omega_0 \neq 0$, then the fundamental period T_0 of $x(t)$ is

$$T_0 = \frac{2\pi}{|\omega_0|} \quad \text{(the smallest positive value of } T\text{)}$$

The signals $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$ have the same fundamental period.

Consider a sinusoidal signal

$$x(t) = A\cos(\omega_0 t + \phi)$$

$$\omega_0 = 2\pi f_0 \quad \text{(fundamental frequency)}$$

The sinusoidal signal, $A\cos(\omega_0 t + \phi)$, is closely related to the periodic complex exponential signal, $e^{j\omega_0 t}$. [Euler's identity]

● Euler's Identity

$$e^{j\omega_0 t} = \cos\,\omega_0 t + j\sin\,\omega_0 t$$

Similarly,

$$A\cos(\omega_0 t+\phi) = \frac{A}{2}e^{j\phi}e^{j\omega_0 t} + \frac{A}{2}e^{-j\phi}e^{-j\omega_0 t}$$

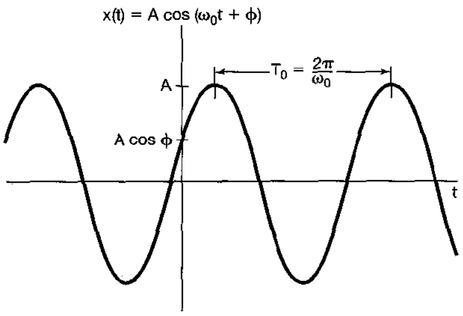
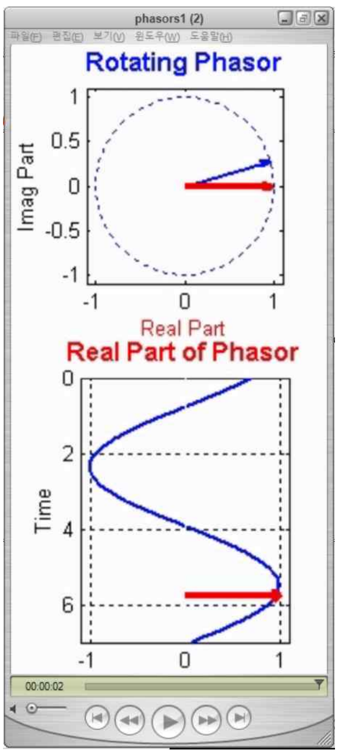
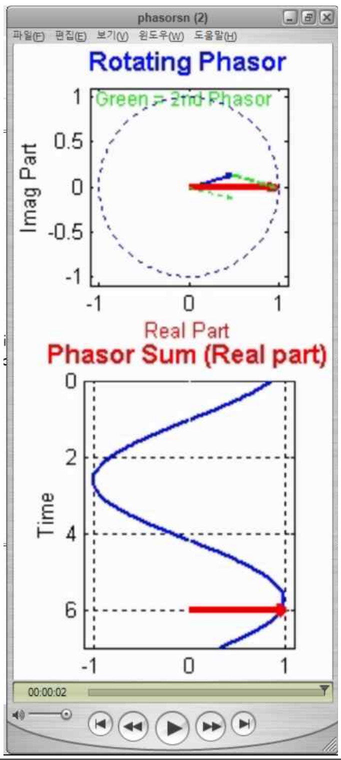


Fig. 20 continuous-time sinusoidal signal

※ How to make cosine waves using phasors

<table>
<tr><td>$\cos(\omega t) = Re\{e^{j\omega t}\}$</td><td>$\cos(\omega t) = \dfrac{e^{j\omega t}+e^{-j\omega t}}{2}$</td></tr>
<tr><td>
</td><td>
</td></tr>
</table>

https://dspfirst.gatech.edu/chapters/02sines/demos/phasors/index.html

(3) <u>General Complex Exponential Signals</u>

- Continuous-time complex exponential signal

$$x(t) = C e^{at} \tag{1.20}$$

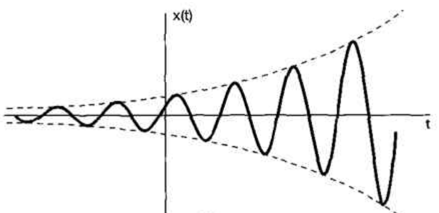
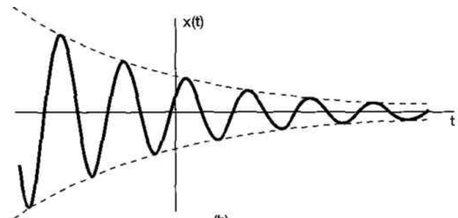
C and a are, in general, complex numbers.

The complex numbers C and a are represented as following:

$$C = |C|\, e^{j\theta} \quad \text{and} \quad a = r + j\omega_0$$

Then,

$$\begin{aligned} x(t) &= C e^{at} = |C|\, e^{j\theta}\, e^{(r+j\omega_0)t} = |C|\, e^{rt}\, e^{j(\omega_0 t+\theta)} \\ &= |C|\, e^{rt} \cos(\omega_0 t + \theta) + j\, |C|\, e^{rt} \sin(\omega_0 t + \theta) \end{aligned}$$

growing sinusoidal signal	decaying sinusoidal signal
$x(t) = C e^{rt} \cos(\omega_0 t + \theta),\ r > 0$	$x(t) = C e^{rt} \cos(\omega_0 t + \theta),\ r < 0$
 (a)	 (b)
※ unstable systems (The poles of a system are on the right half s-plane.)	the response of RLC circuits the automotive suspension systems

※ Sinusoidal signals multiplied by decaying exponentials are commonly referred to as <u>damped sinusoids</u>.

※ <u>Summary</u>

$$x(t) = C\,e^{at}$$

C	a	example	real part
real	real	$5\,e^{-2t}$	$5\,e^{-2t}$
real	complex	$5\,e^{(-2+j3)t}$	$5\,e^{-2t}\cos(3t)$
complex	real	$(5+j5)\,e^{-2t}$	$5\,e^{-2t}$
complex	complex	$(5+j5)\,e^{(-2+j3)t}$	$5\sqrt{2}\,e^{-2t}\cos(3t+45^\circ)$
(*) real	pure imaginary	$5\,e^{j3t}$	$5\cos(3t)$
(*) real	0	$5\,e^{0t}$	5

§1.3.2 Discrete-time Complex Exponential and Sinusoidal Signals

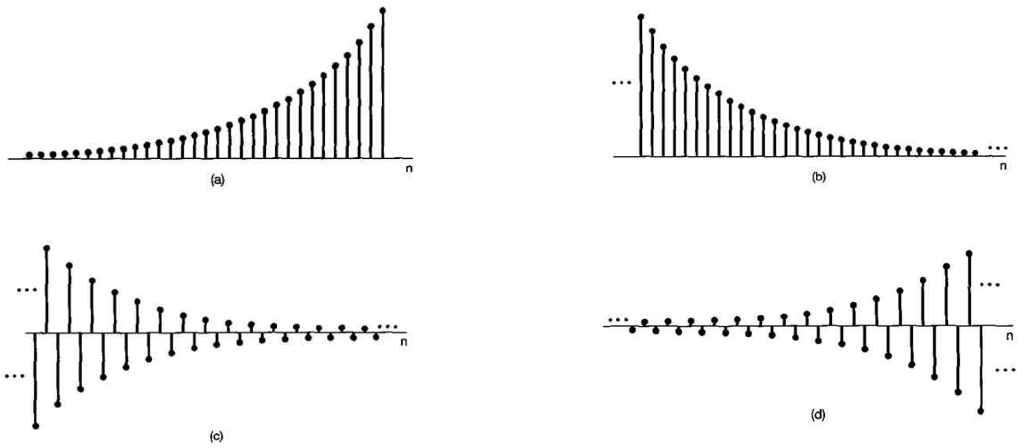
- Discrete-time complex exponential signal (or sequence)

$$x[n] = C\alpha^n \quad \text{or} \quad x[n] = C e^{\beta n} \tag{1.44}$$

C and α are, in general, complex numbers.

(1) Real Exponential Signals

$$x[n] = C\alpha^n, \quad (a)\ \alpha > 1\ ;\ (b)\ 0 < \alpha < 1\ ;\ (c)\ -1 < \alpha < 0\ ;\ (d)\ \alpha < -1$$



(2) Periodic Complex Exponential and Sinusoidal Signals

β is purely imaginary

$$x[n] = e^{j\omega_0 n}$$

Periodic Complex Exponential and Sinusoidal Signals

	periodic complex exponential signal	sinusoidal signal
discrete-time	$x[n] = e^{j\omega_0 n}$	$x[n] = A\cos(\omega_0 n + \phi)$
continuous-time	$x(t) = e^{j\omega_0 t}$	$x(t) = A\cos(\omega_0 t + \phi)$

Euler's Identity

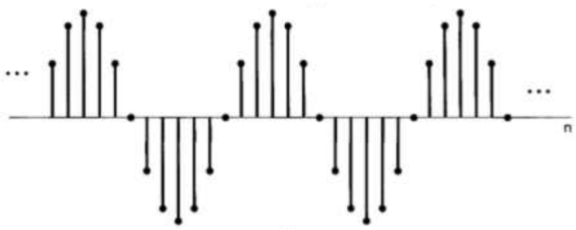
$$e^{j\omega_0 n} = \cos \omega_0 n + j \sin \omega_0 n$$

Similarly,

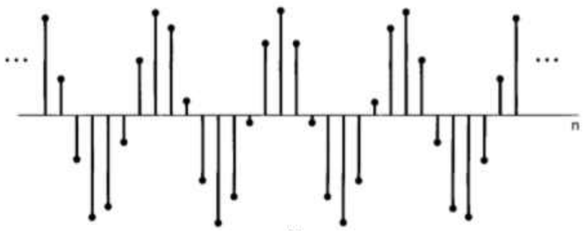
$$A \cos(\omega_0 n + \phi) = \frac{A}{2} e^{j\phi} e^{j\omega_0 n} + \frac{A}{2} e^{-j\phi} e^{-j\omega_0 n}$$

(Example) the discrete-time sinusoidal signals [Fig. 1.25]

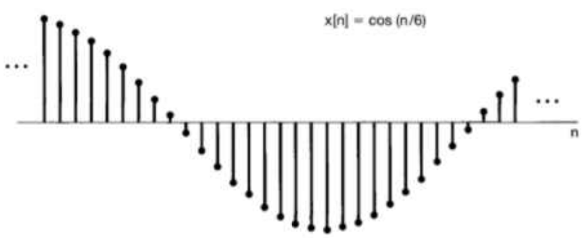
(i) $x[n] = \cos(2\pi n/12)$



(ii) $x[n] = \cos(8\pi n/31)$



(iii) $x[n] = \cos(n/6)$



(3) General Complex Exponential Signals

- Discrete-time complex exponential signal (or sequence)

$$x[n] = C\,\alpha^n \quad \text{or} \quad x[n] = C\,e^{\beta n} \qquad (1.44)$$

C and α are, in general, complex numbers.

The complex numbers C and a are represented as following:

$$C = |C|\,e^{j\theta} \quad \text{and} \quad \alpha = |\alpha|\,e^{j\omega_0}$$

Then,

$$x[n] = C\,\alpha^n = |C|\,|\alpha|^n \cos(\omega_0 n + \theta) + j\,|C|\,|\alpha|^n \sin(\omega_0 n + \theta)$$

$|\alpha| = 1$: sinusoidal

$|\alpha| < 1$: sinusoidal sequences multiplied by a decaying exponential

$|\alpha| > 1$: sinusoidal sequences multiplied by a growing exponential

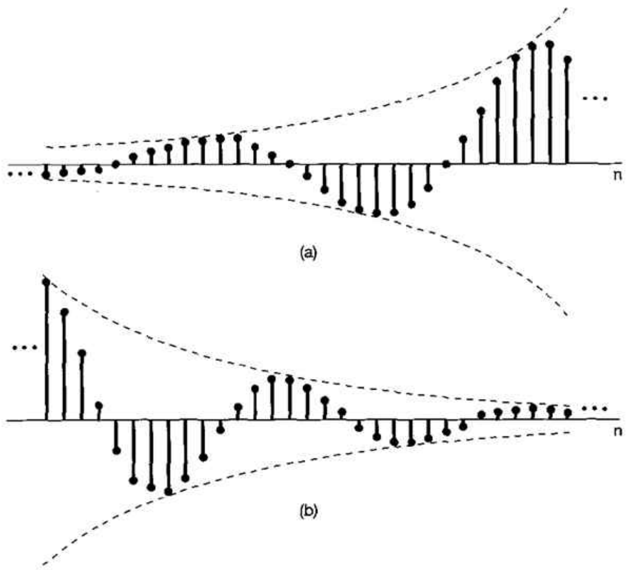


Figure 1.26 (a) Growing discrete-time sinusoidal signals; (b) decaying discrete-time sinusoid.

§1.3.3 Periodicity Properties of Discrete-time Complex Exponentials

[※ Please, read CAREFULLY this section of the textbook.]

While there are many similarities between continuous-time and discrete-time signals, there are also many important differences.

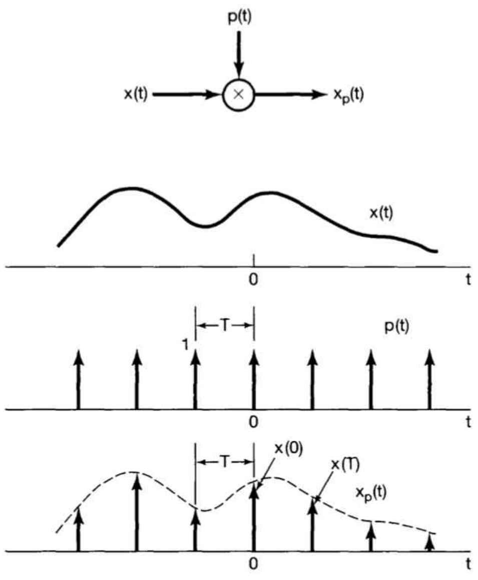
- One of these concerns the discrete-time exponential signal $e^{j\omega_0 n}$.

continuous-time signal $e^{j\omega_0 t}$	discrete-time signal $e^{j\omega_0 n}$
$\omega_0 \uparrow \Rightarrow$ the rate of oscillation $\uparrow$	?
periodic for any value of ω_0	?

Right or Wrong?

discrete	$e^{j(\omega_0+2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = 1\ e^{j\omega_0 n} = e^{j\omega_0 n}$	$e^{j2\pi n} = 1$?
continuous	$e^{j(\omega_0+2\pi)t} = e^{j2\pi t} e^{j\omega_0 t} = 1\ e^{j\omega_0 t} = e^{j\omega_0 t}$	$e^{j2\pi t} = 1$?

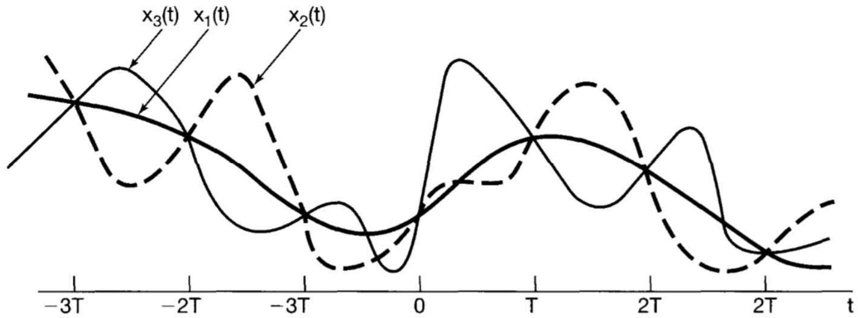
[Sampling] continuous signal $\Rightarrow$ discrete signal



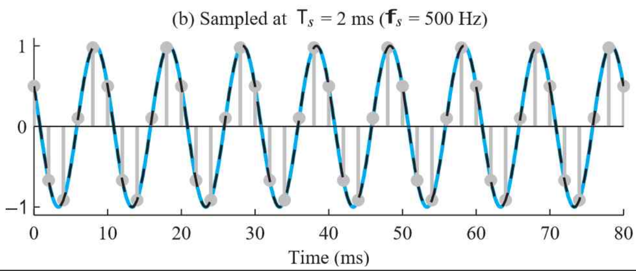
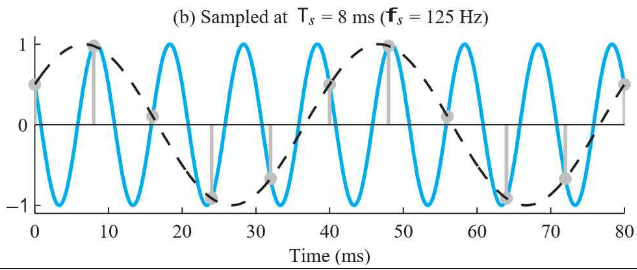
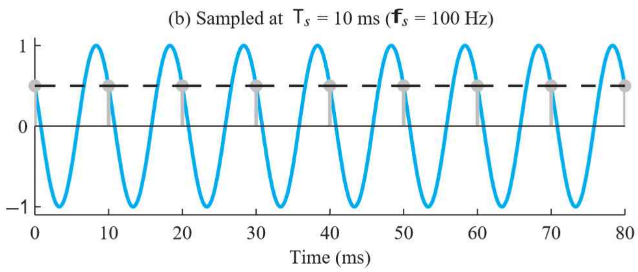
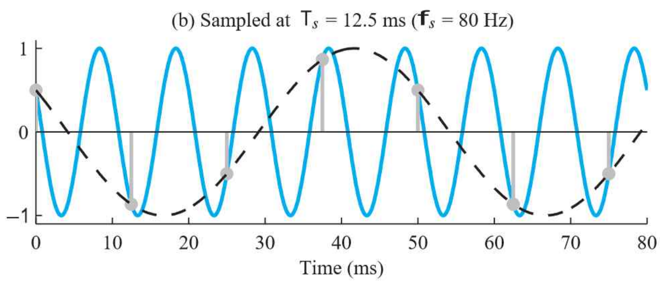
https://www.youtube.com/watch?v=jHS9JGkEOmA

※ 3 continuous time signals $x_1(t) = x_2(t) = x_3(t)$? different signals

3 sampled signals $x_1(kT) = x_2(kT) = x_3(kT)$? same signals

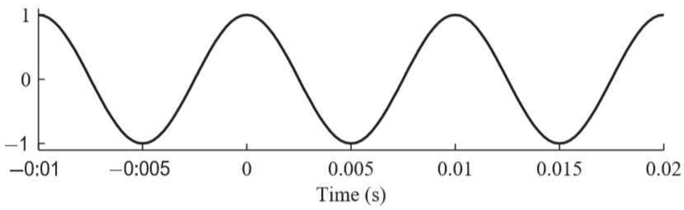


- sampling with different sampling frequencies

$f = 100$ Hz $f_s = 500$ Hz $(T_s = 2\ \mathrm{ms})$ $\omega_0 = \dfrac{2\pi}{5} = 0.4\pi$	
$f = 100$ Hz $f_s = 125$ Hz $(T_s = 8\ \mathrm{ms})$ $\omega_0 = \dfrac{8\pi}{5} = 1.6\pi$ $(= 0.4\pi)$	
$f = 100$ Hz $f_s = 100$ Hz $(T_s = 10\ \mathrm{ms})$ $\omega_0 = 2\pi$ $(= 0\pi)$	
$f = 100$ Hz $f_s = 80$ Hz $(T_s = 12.5\ \mathrm{ms})$ $\omega_0 = \dfrac{10\pi}{4} = 2.5\pi$ $(= 0.5\pi)$	

※ Sampled signal and Digital frequency

original continuous signal $x(t) = \cos(2\pi 100\, t)$ $f = 100\,\text{Hz}$



- original continuous signal $x(t) = \cos(2\pi 100\, t)$ $f = 100\,\text{Hz}$
- sampled discrete signal

$f_s = 2000$ samples/sec (i.e., sampling frequency $f_s = 2000$ Hz)

$T_s = 0.5\,\text{ms}$ (sampling interval)

$x(n\,T_s) = \cos(2\pi 100\; n\,T_s) = \cos(2\pi\,(100\,\text{Hz})\; n\,(0.5\text{ms}))$

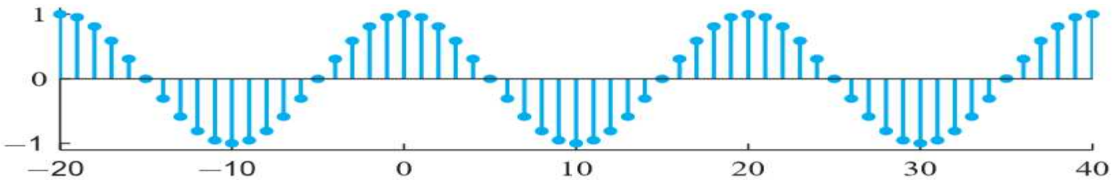
$= \cos(2\pi(\frac{100\,\text{Hz}}{2000\,\text{Hz}})\; n) = \cos(\frac{2\pi}{20}\; n) = \cos(0.1\pi n)$ [No T_s]

$x[n] = \cos(0.1\pi\, n) = \cos(\omega_o n) = Re[e^{j\omega_o n}]$ (※ $\omega_o = 0.1\pi$ rad)

20 samples / period ($= f_s / f = 2000/100$)

period N (instead of T) $= \dfrac{f_s}{f} = \dfrac{2000}{100} = 20$

(20 samples per continuous time signal period)



- original continuous signal $x(t) = \cos(2\pi 100\, t)$ $f = 100\,\text{Hz}$
- sampled discrete signal

$f_s = 500$ samples/sec (i.e., sampling frequency $f_s = 500$ Hz)

$T_s = 2\,\text{ms}$ (sampling interval)

$x(n\,T_s) = \cos(2\pi 100\; n\,T_s) = \cos(2\pi\,(100\,\text{Hz})\; n\,(2\text{ms}))$

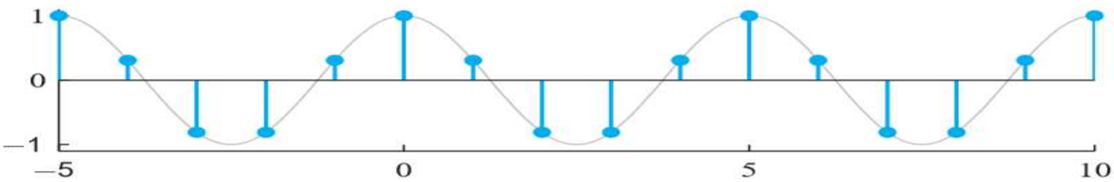
$= \cos(2\pi(\frac{100\,\text{Hz}}{500\,\text{Hz}})\; n) = \cos(\frac{2\pi}{5}\; n) = \cos(0.4\,\pi\, n)$ [No T_s]

$x[n] = \cos(0.4\,\pi\, n) = \cos(\omega_o n) = Re[e^{j\omega_o n}]$ (※ $\omega_o = 0.4\pi$ rad)

5 samples / period ($= f_s / f = 500/100$)

period N (instead of T) $= \dfrac{f_s}{f} = \dfrac{500}{100} = 5$

(5 samples per continuous time signal period)



※ analog frequency / sampling frequency / digital frequency

frequency of original continuous time signal f	sampling frequency f_s	digital frequency of the sampled signal $\omega_o = 2\pi(\frac{f}{f_s})$
100 Hz	2,000 Hz	0.1π rad
200 Hz	4,000 Hz	0.1π rad
1,000 Hz	20,000 Hz	0.1π rad
10,000 Hz	200,000 Hz	0.1π rad
bigger original freq.		same digital freq.

※ analog frequency / sampling frequency / digital frequency

frequency of original continuous time signal f	sampling frequency f_s	digital frequency of the sampled signal $\omega_o = 2\pi(\frac{f}{f_s})$
100 Hz	2,000 Hz	0.1π rad
100 Hz	1,000 Hz	0.2π rad
100 Hz	500 Hz	0.4π rad
100 Hz	400 Hz	0.5π rad
same original freq.		bigger digital freq.

[** Important **]

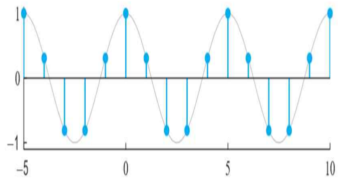
continuous-time signal $e^{j\omega_0 t}$	ω_o [rad/sec] **is the real frequency.**
discrete-time signal $e^{j\omega_0 n}$	ω_o [rad] **is not the real frequency, but the sampling interval in radian.**

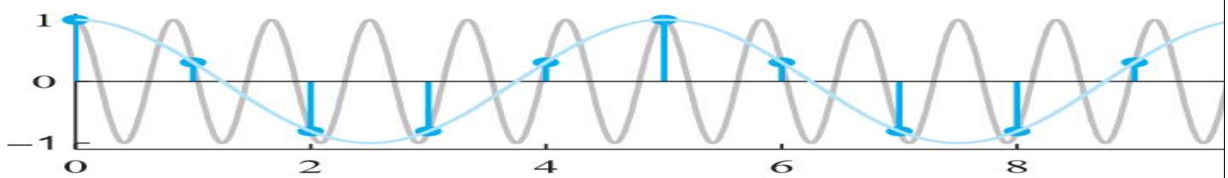
● $\cos(0.4\pi n) = \cos(2.4\pi n)$?

- original continuous signal $x(t) = \cos(2\pi 100\, t)$ $f = 100$ Hz
- sampled discrete signal

$T_s = 2$ms (sampling interval) (i.e., sampling frequency $f_s = 500$ Hz)

$$x(nT_s) = \cos(2\pi 100\, n T_s) = \cos(2\pi (100\text{Hz})\, n\, (2\text{ms}))$$
$$= \cos\left(2\pi \left(\frac{100\,\text{Hz}}{500\,\text{Hz}}\right) n\right) = \cos\left(\frac{2\pi}{5}\, n\right) = \cos(0.4\pi n) \quad [\text{ No } T_s \,]$$
$$x[n] = \cos(0.4\pi n) = \cos(\omega_o n) = Re[e^{j\omega_o n}] \quad (\text{※ } \omega_o = 0.4\pi \text{ rad})$$





- original continuous signal $x(t) = \cos(2\pi 100\, t)$ $f = 100$ Hz
- sampled discrete signal

$T_s = 12$ms (sampling interval) (i.e., sampling frequency $f_s = 83.333$ Hz)

$$x(nT_s) = \cos(2\pi 100\, n T_s) = \cos(2\pi (100\text{Hz})\, n\, (12\text{ms}))$$
$$= \cos(2.4\pi n)$$
$$x[n] = \cos(2.4\pi n) = \cos(\omega_o n) = Re[e^{j\omega_o n}] \quad (\text{※ } \omega_o = 2.4\pi \text{ rad})$$

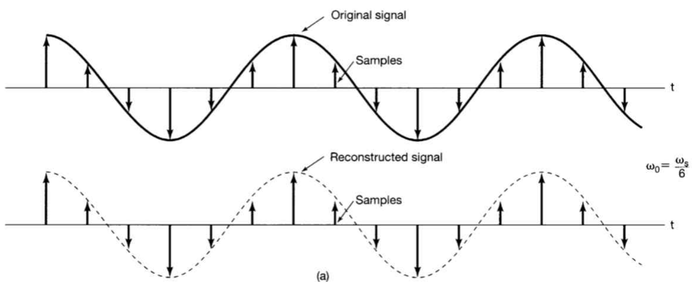
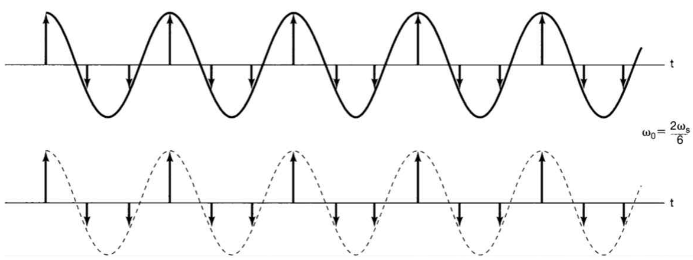
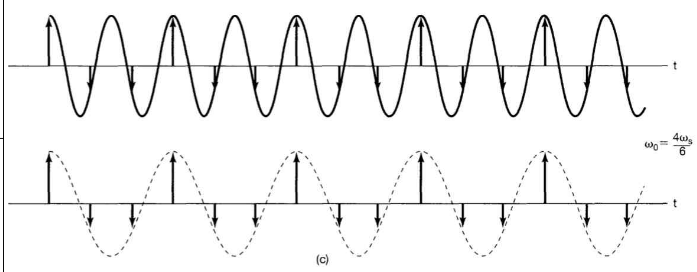
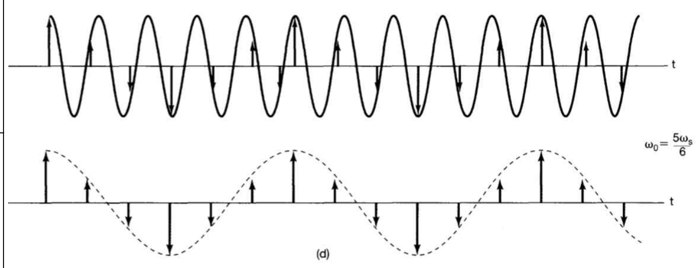
※ $e^{j2.4\pi n} = e^{j(2+0.4)\pi n} = e^{j2\pi n}\, e^{j0.4\pi n} = 1\, e^{j0.4\pi n} = e^{j0.4\pi n}$

$$e^{j(\omega_0 + 2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = e^{j\omega_0 n}$$
$$e^{j(\omega_0 + 4\pi)n} = e^{j4\pi n} e^{j\omega_0 n} = e^{j2\pi n} e^{j2\pi n} e^{j\omega_0 n} = e^{j\omega_0 n}$$

So, these digital frequencies are the same frequencies.

$$\omega_0 = \omega_0 \pm 2\pi = \omega_0 \pm 4\pi = \cdots = \omega_0 \pm (2\pi)n = \cdots$$

※ reconstructed signal using sampled signal

original $\omega_0 = \frac{2\pi}{6} = \frac{\pi}{3}$	
reconstructed $\omega_0 = \frac{2\pi}{6} = \frac{\pi}{3}$	
original $\omega_0 = \frac{2\pi}{3}$	
reconstructed $\omega_0 = \frac{2\pi}{3}$	
original $\omega_0 = \frac{4\pi}{3}$	
reconstructed $\omega_0 = \frac{2\pi}{3}$	
original $\omega_0 = \frac{10\pi}{6} = \frac{5\pi}{3}$	
reconstructed $\omega_0 = \frac{2\pi}{6} = \frac{\pi}{3}$	

※ $\omega_0 \uparrow$ ⇒ the rate of oscillation ↑ in the continuous-time domain?

※ $\omega_0 \uparrow$ ⇒ the rate of oscillation ↑ in the discrete-time domain?

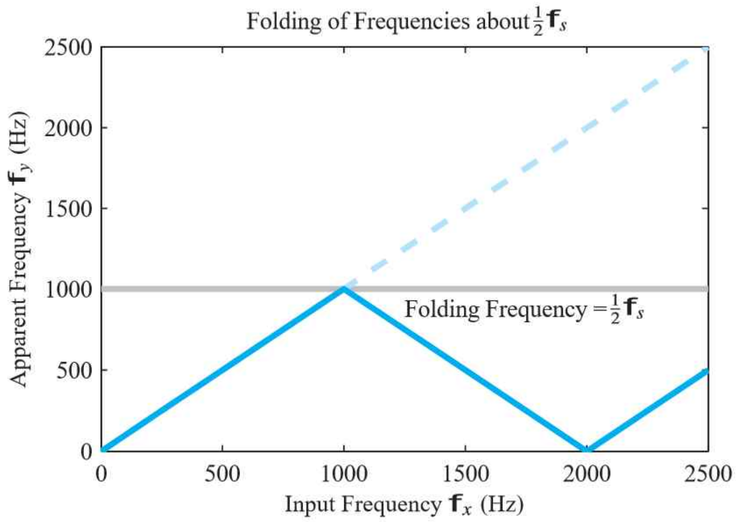
1. Since $e^{j(\omega_0+2\pi)n} = e^{j2\pi n}e^{j\omega_0 n} = e^{j\omega_0 n}$ in the discrete-time case, the signal with frequency ω_0 is identical to the signals with frequencies $\omega_0 \pm 2\pi$, $\omega_0 \pm 4\pi$, and so on. Therefore, in considering discrete-time complex exponentials, we need only consider a frequency interval of length 2π ($0 \leqq \omega_0 < 2\pi$ or $-\pi \leqq \omega_0 < \pi$).
   [※ This fact is also TRUE in the continuous-time case?]

2. The signal $e^{j\omega_0 n}$ does not have a continually increasing rate of oscillation as ω_0 is increased in magnitude. (Fig. 1.27)

   $\omega_0 : 0 \rightarrow \pi$ The rate of oscillation is increased.

   $\omega_0 : \pi \rightarrow 2\pi$ The rate of oscillation is decreased.

   (the revolutions of the phasor, aliasing)

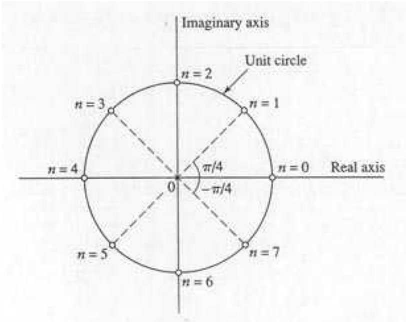


sampling frequency = 2000 Hz

https://www.youtube.com/watch?v=VNftf5qLpiA
https://www.youtube.com/watch?v=ByTsISFXUoY
https://www.youtube.com/watch?v=B8EMI3_0TO0

(Example) Complex exponential $e^{j\omega_0 n}$ for $\omega_0 = \pi/4$

$$\omega_0 = \frac{\pi}{4} \text{ and } n = 0, 1, 2, \cdots, 7, [8, 9, 10, \cdots] \quad \text{[Fig. 1.27(c)]}$$



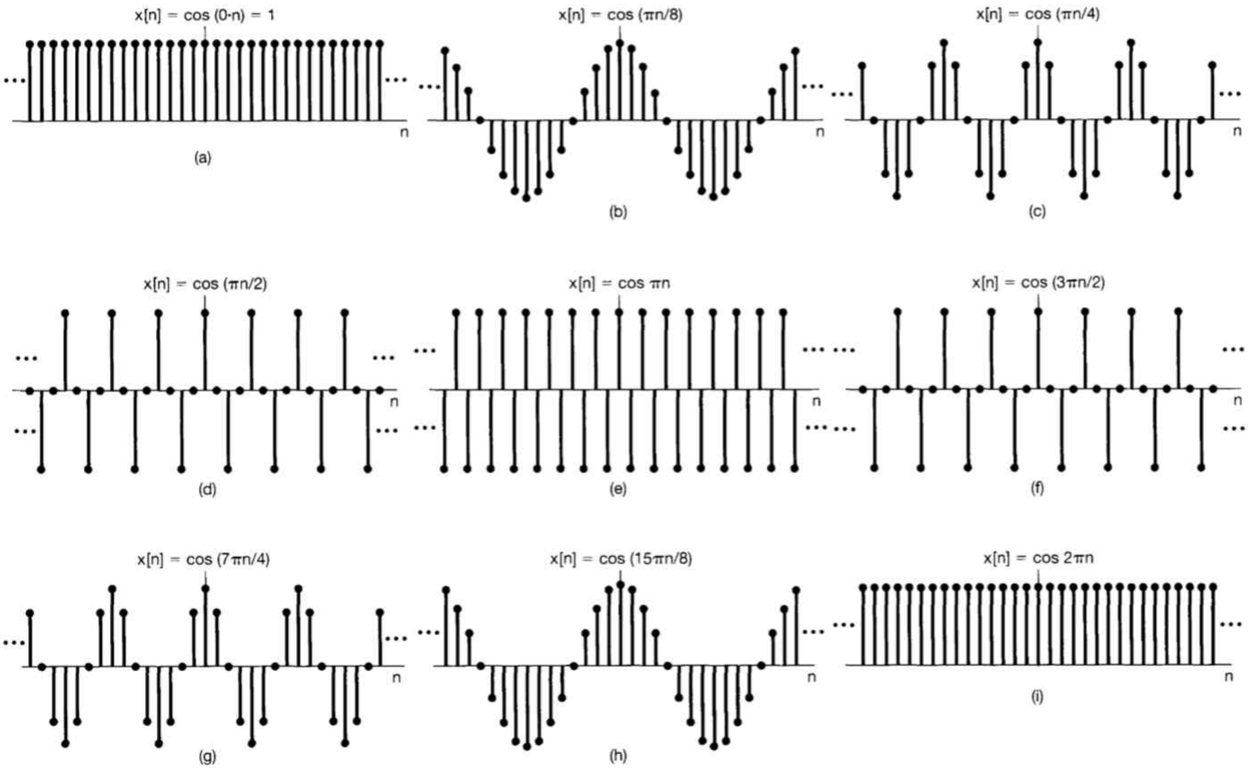
ω_0 near $0, \pm 2\pi, \pm 4\pi, ...$: low frequency (slow variation)

ω_0 near $\pm\pi, \pm 3\pi, \pm 5\pi, ...$: high frequency (rapid variation)

For $\omega_0 = \pi$ or odd multiples of π, $e^{j\pi n} = (e^{j\pi})^n = (-1)^n$

⇒ oscillates rapidly (changing sign at each point in time) [Fig. 1.27(e)])

※ Discrete time sinusoidal sequences for several frequencie



- Periodic for any value of ω_0 in the discrete-time domain?

the signal $e^{j\omega_0 n}$ to be PERIODIC with period $N > 0$	$\Rightarrow$	$e^{j\omega_0 (n+N)} = e^{j\omega_0 n}$ or $e^{j\omega_0 N} = 1$ or $\omega_0 N =$ a multiple of 2π

There must be an integer m such that $\omega_0 N = 2\pi m$.

So, if $\dfrac{\omega_0}{2\pi} = \dfrac{m}{N}$ is a rational number, $e^{j\omega_0 n}$ is periodic.

(Example)

Signals shown in Fig. 1.25(a) and (b) : periodic

Signals shown in Fig. 1.25(c) : aperiodic

※ Comparison of the signals $e^{j\omega_0 t}$ and $e^{j\omega_0 n}$

	$e^{j\omega_0 t}$	$e^{j\omega_0 n}$
fundamental period	T [sec]	N [no unit]
fundamental (angular) frequency	$\omega_0 = \dfrac{2\pi}{T}$ [rad/sec]	$\omega_0 = \dfrac{2\pi}{N}$ [rad]
fundamental frequency	$f_0 = \dfrac{1}{T}$ [Hz or 1/sec]	$f_0 = ?$ (undefined)
periodic for	any choice of ω_0	$\omega_0 = \dfrac{2\pi m}{N}$
remarks	Distinct signals for distinct values of ω_0	Identical signals for values of ω_0 separated by multiples of 2π

※ Physical meaning of ω_0 [rad/sec] in the continuous-time domain?

※ Physical meaning of ω_0 [rad] in the discrete-time domain?

continuous-time signal $e^{j\omega_0 t}$	ω_o [rad/sec] **is the real frequency.**
discrete-time signal $e^{j\omega_0 n}$	ω_o [rad] **is not the real frequency, but the sampling interval in radian.**

(Example) Periodic/Nonperiodic (Fig. 1.25)

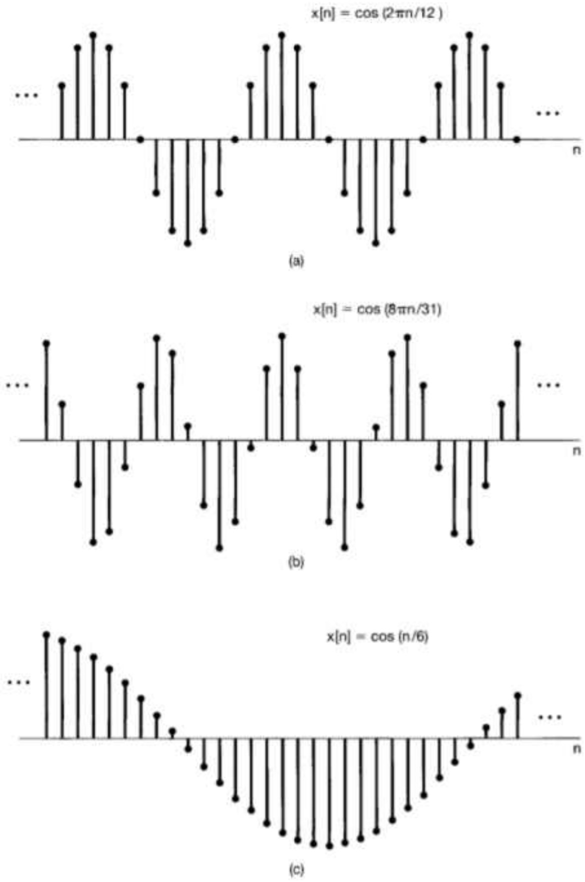


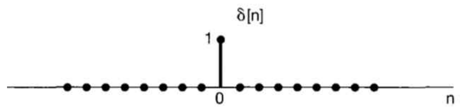
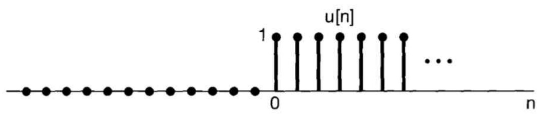
Figure 1.25 Discrete-time sinusoidal signals.

	Fig. 1.25(a)	Fig. 1.25(b)	Fig. 1.25(c)
$x[n]$	$\cos(\frac{2\pi n}{12})$	$\cos(8\pi n/31)$	$\cos(n/6)$
ω_0	$\frac{2\pi}{12}$	$\frac{8\pi}{31}=\frac{2\pi}{31/4}$	$\frac{1}{6}=\frac{2\pi}{12\pi}$
$\frac{N}{m}$	12 (integer)	$\frac{31}{4}$ (rational)	12π (irrational)
repeat every ...	12 points (= one cycle)	31 points (= four cycles)	12π points (?)
theoretically periodic?	YES	YES	NO
periodic in real?	YES	YES	?

1.4 The Unit Impulse and Unit Step Functions

1.4.1 The Discrete-time Unit Impulse and Unit Step Sequences

- Basic discrete-time signals

discrete-time unit impulse (or unit sample)	discrete-time unit step
$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$	$u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geqq 0 \end{cases}$
 Fig. 1.28	 Fig. 1.29

$\delta[n]$ is the first difference of $u[n]$

$$\delta[n] = u[n] - u[n-1]$$

First difference (※ differential / difference)

(Reference)

1. $u[n] = \sum_{m=-\infty}^{n} \delta[m]$ (※ $u[n]$ is the running sum of $\delta[n]$.)

$$= \sum_{k=0}^{\infty} \delta[n-k]$$

2. Sampling property

$$x[n]\,\delta[n] = x[0]\,\delta[n]$$

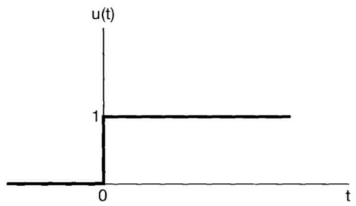
$$x[n]\,\delta[n-n_0] = x[n_0]\,\delta[n-n_0]$$

(It will play an important role in Chapter 2.)

1.4.2 The Continuous-time Unit Step and Unit Impulse Function

- Continuous-time unit step function $u(t)$

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



- Continuous-time unit impulse $\delta(t)$

(≡ the first derivative of $u(t)$)

$$\delta(t) = \frac{du(t)}{dt}$$

※ Considering an approximation

$$\delta(t) = \lim_{\Delta \to 0} \delta_{\Delta}(t)$$

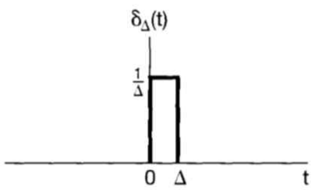


Figure 1.34 Derivative of $u_\Delta(t)$.

⇒

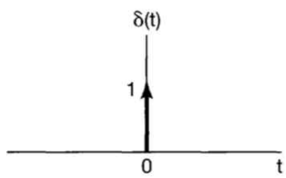


Figure 1.35 Continuous-time unit impulse.

$\delta[n]$: '1' means the magnitude. [Fig. 1.35]

$\delta(t)$: '1' means the area. [Fig. 1.28]

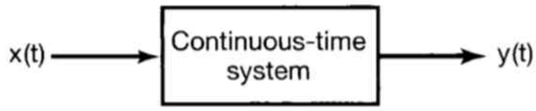
※ Singularity functions

(Reference)

1. $u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau = \int_{0}^{\infty} \delta(t-\sigma)\, d\sigma$
2. $x(t)\,\delta(t) = x(0)\,\delta(t)$
   $x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$

1.5 Continuous-time and Discrete-time Systems

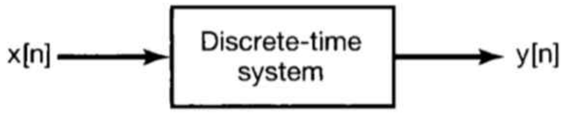
- Continuous-time system



First-order linear differential equation

$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

- Discrete-time system

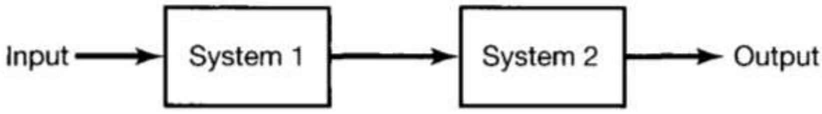


First-order linear difference equation

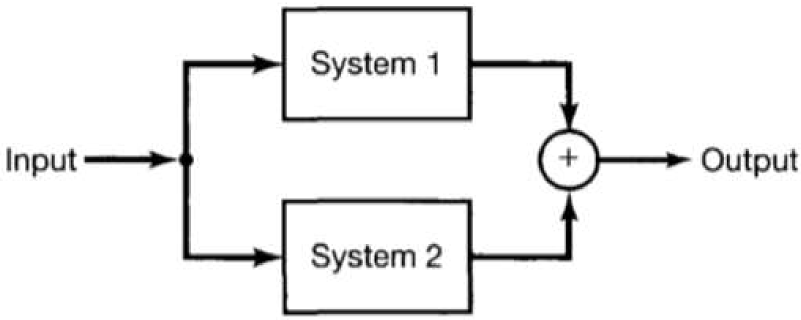
$$y[n] + ay[n-1] = bx[n]$$

- Interconnections of systems

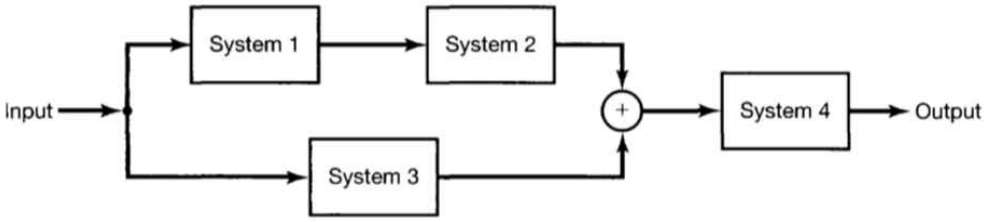
1. series (cascade) interconnection



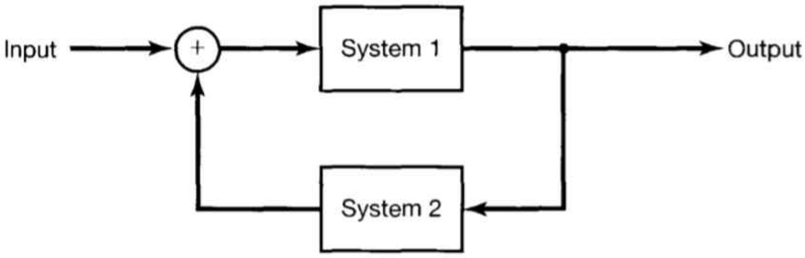
2. parallel interconnection



3. series-parallel interconnection



4. feedback interconnection



1.6 Basic System Properties

1.6.1 Systems with and without Memory

- Systems without memory (memoryless systems)

{ current input } ⇒ current output

(Example) resistor, $y(t) = R\,x(t)$, $y[n] = 2x[n]$

- Systems with memory

$$\left\{\begin{array}{r} \text{current input} \\ \text{past input} \\ \text{future input} \end{array}\right\} \Rightarrow \text{current output}$$

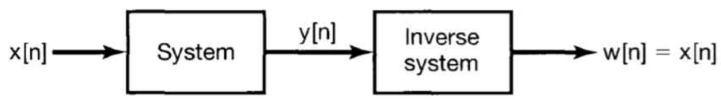
(Example)

accumulator ($y[n] = \sum_{k=-\infty}^{n} x[k]$), delay($y[n] = x[n-1]$),

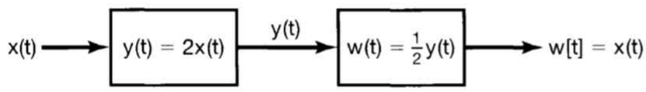
capacitor ($y(t) = \frac{1}{C}\int_{-\infty}^{t} x(\tau)\,d\tau$), sequential circuit,

feedback system ($y[n] = y[n-1] + x[n]$)

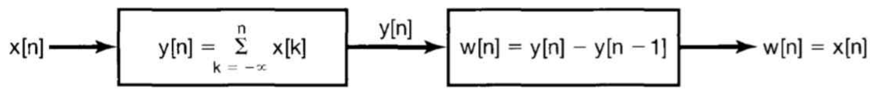
1.6.2 Invertibility and Inverse Systems



(Example)



(b)



- Transfer function

poles of a system ⇒ zeros of the inverse system

zeros of a system ⇒ poles of the inverse system

∴ The inverse system of a system does not exist in some case.

(Example)

Transmitter/receiver [noise? → probability (ex) blind equalizer]

Modulator/demodulator (modem)

Encoder/decoder [lossless coding (ex) jpg file(lossy or lossless?)]

1.6.3 Causality

A system is causal if the output at any time depends on values of the input at only the PRESENT and PAST times.

(Examples)

Causal : $y[n] = x[n] + x[n-1]$, all memoryless systems

Noncausal : $y[n] = x[n] + x[n+1]$, $y[n] = \dfrac{1}{2M+1}\sum_{k=-M}^{M} x[n-k]$

Ideal systems : noncausal systems

1.6.4 Stability

Bounded Input Bounded Output (BIBO) stable

Examples: simple RC circuit

Stability will be studied in detail in the Control Theory class.

1.6.5 Time Invariance

$$x(t) \rightarrow \boxed{\text{system}} \rightarrow y(t) \rightarrow \boxed{\text{delay } t_0} \rightarrow y(t-t_0)$$

$$x(t) \rightarrow \boxed{\text{delay } t_0} \rightarrow x(t-t_0) \rightarrow \boxed{\text{system}} \rightarrow y(t-t_0)$$

1.6.6 Linearity

$$x(t) \rightarrow \boxed{\text{system}} \rightarrow y(t)$$

1. Homogeneity

$$a\,x(t) \rightarrow \boxed{\text{system}} \rightarrow a\,y(t) \quad \text{for any complex constant } a$$

2. Additivity

$$x_1(t) + x_2(t) \rightarrow \boxed{\text{system}} \rightarrow y_1(t) + y_2(t)$$

(homogeneity + additivity)

Continuous time : $a\,x_1(t) + b\,x_2(t) \rightarrow a\,y_1(t) + b\,y_2(t)$

Discrete time : $a\,x_1[n] + b\,x_2[n] \rightarrow a\,y_1[n] + b\,y_2[n]$

※ Linear Combination

(input) $$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + ...$$

(output) $$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + ...$$

§1.7 Summary

Superposition

Linear combination

Series (Taylor series, Fourier series, Maclaurin series,
power series, ...), polynomial, vector, Euler's identity, ...

Orthogonality

Divide and conquer (simplify)

Linear system

Time invariant system

Linear time-invariant (LTI) system